

SMITHIAN FOLD THEORY - REVERSIBLE AND QUANTUM COMPUTATION BRANCH PAPER 001

The Quantum Fold Machine

An Exact, Parameter-Free and Machine-Closed Derivation of Reversible and
Quantum Computation from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

Third clean-room reconstruction - Quantum Computation inventory complete

21 admitted derivations - 5,376 generated candidates

23 July 2026

Paper: CC BY 4.0 - Code: Apache-2.0

LOCAL PREPUBLICATION MANUSCRIPT - PUBLICATION NOT YET AUTHORIZED

Abstract

This paper reports the completed Reversible and Quantum Computation branch of the third clean-room reconstruction of Smithian Fold Theory. From the admitted Foundation, Mathematics, Information Science and Classical Computation receipts it derives a complete reversible model; Fold quantum information units and composition; superposition-equivalent support; phase and interference; entanglement; measurement; gates and circuits; quantum universality, algorithms and complexity; communication; coding, error correction and fault tolerance; simulation; verification; learning; full operational classical-quantum correspondence; and the limits of quantum computation. It imports no complex amplitude, Hilbert-space axiom, stochastic collapse postulate, fitted parameter or physical benchmark as a premise.

The frozen branch contains 21 dependency-ordered claims and 5,376 generated candidate structures. Every candidate is decided, each grammar has exactly one survivor, and every claim passes minimality, named-shape uniqueness, depth-independent closure, four adverse controls, cryptographic sealing and implementation-distinct recomputation.

Inventory identity:

sha256:8cb45670ff6fd49ede15abd8ca03268438de5d100ef70beab0d9ca42b78e8141.

Keywords: Smithian Fold Theory; reversible computation; quantum information; superposition; phase; interference; entanglement; measurement; quantum circuits; error correction; fault tolerance; quantum algorithms; quantum complexity; open computational science

1. Central scientific claim and exact boundary

Within the frozen SFT V3 Reversible and Quantum Computation inventory, all 21 registered obligations have depth-independent, model-admitted and independently replicated receipts, and no registered quantum-computation obligation remains unclassified or frontier.

Closure is formal and structural. It does not by itself claim that a physical device realizes these operations, supply a measured fault threshold, establish unrestricted speedups, or derive quantum physics as a natural law. Those require the later empirical Physics branch.

2. Fold-native quantum constitution

One Fold distinction supplies the native information unit. Complete held-label words across generated positions supply branch support. A branch carries an exact held phase label from a generated finite cycle. Joint states are complete pair-cell words retaining component identity. A transformation is a total branch-word bijection plus a period-held phase action. None of these definitions requires a real or complex amplitude, irrational normalization, imaginary proof value or ungenerated continuum.

3. Superposition, interference and entanglement

Superposition-equivalent means complete simultaneous support of every generated branch before an observation class is retained; it does not install an ontic stochastic mixture. Phase is a reversible action on held cycle labels. Interference is a complete branch-to-image relation whose predecessor fibres preserve every merging branch and its phase. Joint support is factorable exactly when it equals the Cartesian product of its marginals; a strict correlated subset is nonfactorable and supplies the entangling class.

4. Measurement and reversibility

Measurement is a total observation relation over complete branch support. The selected observation class becomes the retained state, while the measurement record keeps every pre-observation branch, phase and image label. Support reduction is therefore exact and reconstructible at the record boundary. A gate is reversible only when its branch map is bijective and its phase action has an exact inverse; any predecessor merge requires a retained predecessor label.

5. Coding, multi-error correction and fault tolerance

A generated fault-depth trace of length t forces repetition width $2t+1$: at least $t+1$ copies retain the source label after any family of at most t label changes. The executable census exhausts every mask through one error at width 3 (4 masks), two errors at width 5 (16 masks), and three errors at width 7 (64 masks). The base and successor certificate extends the majority separation to every generated positive finite t . Fault-tolerant circuits must contain declared faults, prevent one location from opening more distinctions than the code can repair, and recover before errors escape the registered support. No hardware threshold constant is asserted.

6. Classical-quantum correspondence and limits

Classical computation embeds as the single-held-branch, phase-insensitive reversible submodel. Quantum execution extends that carrier with complete branch support, phase-sensitive predecessor merging and nonfactorable joint support. Observation decodes exact classical records. Because every quantum circuit remains a generated finite description executed by an admitted universal process, self-reference and halting boundaries transfer: quantum computation does not acquire an undeclared oracle or hypercomputational escape.

7. Complete execution census

Order	Claim	Candidates	Closure
1	SFT-QUANTUM-REVERSIBLE-MODEL-001	256	depth-independent
2	SFT-QUANTUM-INFORMATION-UNIT-001	256	depth-independent
3	SFT-QUANTUM-STATE-COMPOSITION-001	256	depth-independent
4	SFT-QUANTUM-SUPERPOSITION-001	256	depth-independent
5	SFT-QUANTUM-PHASE-INTERFERENCE-001	256	depth-independent
6	SFT-QUANTUM-ENTANGLEMENT-001	256	depth-independent
7	SFT-QUANTUM-MEASUREMENT-001	256	depth-independent
8	SFT-QUANTUM-GATE-001	256	depth-independent
9	SFT-QUANTUM-CIRCUIT-001	256	depth-independent
10	SFT-QUANTUM-UNIVERSALITY-001	256	depth-independent
11	SFT-QUANTUM-ALGORITHMS-001	256	depth-independent
12	SFT-QUANTUM-COMPLEXITY-001	256	depth-independent
13	SFT-QUANTUM-COMMUNICATION-001	256	depth-independent
14	SFT-QUANTUM-CODING-001	256	depth-independent
15	SFT-QUANTUM-ERROR-CORRECTION-001	256	depth-independent
16	SFT-QUANTUM-FAULT-TOLERANCE-001	256	depth-independent
17	SFT-QUANTUM-SIMULATION-001	256	depth-independent
18	SFT-QUANTUM-VERIFICATION-001	256	depth-independent

Order	Claim	Candidates	Closure
19	SFT-QUANTUM-LEARNING-001	256	depth-independent
20	SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001	256	depth-independent
21	SFT-QUANTUM-LIMITS-001	256	depth-independent

The branch total is **5,376 candidates**, **21 survivors**, **84 adverse controls** and **21 independent validations**.

8. Derivation 1: Complete Fold reversible-computation model

Claim: SFT-QUANTUM-REVERSIBLE-MODEL-001

8.1 Scientific necessity and exact theorem

Complete Fold reversible-computation model must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

8.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for reversible computation.

All finite reversible computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

8.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

8.4 Operational laws and consequences

- Support law: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

8.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact reversible computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

8.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.	PASS

8.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reversible computation, Bennett machine.

This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

8.8 Exact evidence identities

- Source manifest: sha256:c5af094fe52290461356bf39e9d76f13ba2b6202e285508bc50ef96cd4b8e86f
- Complete-census certificate:
sha256:dd9994b76e30d066ca995523aa7528f47f985b73200c8e677750fa7f652c5033
- Derivation seal: sha256:37e9625bca1b99f14969057b963782c6b1cb62efb98ec317178cb32901e2f48c
- Independent implementation:
sha256:088141be2c397e4c3f7e640dbc6ba3850bc936eb3433537b058e95a25e98e2df
- Independent certificate:
sha256:33e527c37a50844de7b1b6228f92b22f4eb6a5279fe576a97b952c28885c5ed1
- External validation:
sha256:ded9f371a7b58164c55628cc9684d6944f8083be860caea6cb43f4f0a629bb60
- Model-admitted engine receipt:
sha256:068b389a38d47c06118600324bc263c951e6e19c6f591f257c601f6125525298
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-REVERSIBLE-MODEL-001-068b389a38d47c06.json

9. Derivation 2: Fold quantum information-unit law

Claim: SFT-QUANTUM-INFORMATION-UNIT-001

9.1 Scientific necessity and exact theorem

Fold quantum information-unit law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

9.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum information units.

All finite quantum information units structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The unit is structural support, not an imported complex vector or conventional qubit.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

9.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

9.4 Operational laws and consequences

- Support law: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

9.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum information units record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

9.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The unit is structural support, not an imported complex vector or conventional qubit.	PASS

9.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: qubit, quantum information unit.

The unit is structural support, not an imported complex vector or conventional qubit.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The unit is structural support, not an imported complex vector or conventional qubit.

9.8 Exact evidence identities

- Source manifest: sha256:a27e2cd1d555c89028a74a2426b74986adf84a83e8fac584985522f90e8b0c83
- Complete-census certificate:
sha256:76b580cc45da482f4fdbd7ad583257f4d3d7719b8062af69b5ac96098cc2f01c
- Derivation seal: sha256:8dd1b7f8a5b144885d8926f1d724e9f1ecc767f71ce3110a1884d03297673cdb
- Independent implementation:
sha256:3a82e4d5098bfd4d453af17516b932f696f816a15c6bc718253472c762d0ee10
- Independent certificate:
sha256:c7ca84cd3a8c47cd8a5e345697f2648d3861372f3c6a72c96a1288ce660f8ef5
- External validation:
sha256:7e138a1d7e2f5ce7784a9eab262e857118f8efce2a96a7ec20b6674541cc60a5
- Model-admitted engine receipt:
sha256:3cae970ebb8336d464bcbc64e17d0f50049da6f018b95f98b53b04a46873bf72
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM- INFORMATION-UNIT-001-3cae970ebb8336d4.json

10. Derivation 3: Fold quantum state-composition law

Claim: SFT-QUANTUM-STATE-COMPOSITION-001

10.1 Scientific necessity and exact theorem

Fold quantum state-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM- INFORMATION-UNIT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001

- SFT - INFO - QUANTUM - CORRESPONDENCE - 001

10.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum state composition.

All finite quantum state composition structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Only finite generated joint supports are admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

10.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.

Eliminated candidates	Exact first-failure reason
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

10.4 Operational laws and consequences

- Support law: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

10.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum state composition record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

10.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.

- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Only finite generated joint supports are admitted.	PASS

10.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: tensor product, composite state.

Only finite generated joint supports are admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Only finite generated joint supports are admitted.

10.8 Exact evidence identities

- Source manifest: sha256:e19a28acfd158a2adc8faa62d1c641983e613fd02abd4f5703fd4df71b5a2e32
- Complete-census certificate:
sha256:d10f6ac72430d0adbf42711c2e21033c6ef6a3fd14d43e9b6aed60ef5336f9d4
- Derivation seal: sha256:07accac25aaa131af96d31bdce98536c3ff39c0c0504a69b146f4a02240441c5
- Independent implementation:
sha256:45b6bcd57a43ea420e30db2d983e862b2206a61b2483270be469b035b5263ee0
- Independent certificate:
sha256:47fa45ac826615121aeec5d2ea41548aebd01dbac38c96e8e2ee8b359cb507e4
- External validation:
sha256:7ac24331062df574d62dbb00a5237cdf8bcd928212a5d0baf95a6cef91b867c1
- Model-admitted engine receipt:
sha256:a358610587ee2c7c6dfcba8a782e46f89a1668e2c86739eac0e8c88d6af719c2

- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-STATE-COMPOSITION-001-a358610587ee2c7c.json

11. Derivation 4: Superposition-equivalent Fold support law

Claim: SFT-QUANTUM-SUPERPOSITION-001

11.1 Scientific necessity and exact theorem

Superposition-equivalent Fold support law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

11.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for superposition-equivalent structure.

All finite superposition-equivalent structure structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No amplitude magnitude, normalization constant or stochastic ontology is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

11.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

11.4 Operational laws and consequences

- Support law: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.

- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

11.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact superposition-equivalent structure record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

11.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No amplitude magnitude, normalization constant or stochastic ontology is imported.	PASS

11.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: superposition, basis support.

No amplitude magnitude, normalization constant or stochastic ontology is imported.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No amplitude magnitude, normalization constant or stochastic ontology is imported.

11.8 Exact evidence identities

- Source manifest: sha256:0d1eb45fe6d4e484dfcaa6ef334a3ab671ddd0da9f6f153b901cc8351f7603e3
- Complete-census certificate:
sha256:95ed3e715d21daabab6325f0d1dbeffdd25e7fdaf0c8c481c2bee0c61bd2e0a7
- Derivation seal: sha256:3d8ee27d90c7ddfea4b88ece679df71f39ba1602da046edef08f8e3ed4a8bcdd
- Independent implementation:
sha256:945ffdb45c2324c21427547d49b7e14fbe31aeaca26deee602a921d72088a911
- Independent certificate:
sha256:0de2881ba9754d71a5ad9d18c7283eafe2a63e31b131dc545c7234b8b668bc63
- External validation:
sha256:5fd70c68356f3013863b86feacd8f2a221a4002161cc5be3074f29e0c6b719e1
- Model-admitted engine receipt:
sha256:54c09712a4a53c16054fc2c8de1e247007b718e1f6f0cd9cd38a14b1a5cbe7d1
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-SUPERPOSITION-001-54c09712a4a53c16.json

12. Derivation 5: Fold phase and interference operation law

Claim: SFT-QUANTUM-PHASE-INTERFERENCE-001

12.1 Scientific necessity and exact theorem

Fold phase and interference operation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-SUPERPOSITION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001

- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

12.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for phase and interference.

All finite phase and interference structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

12.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.

Eliminated candidates	Exact first-failure reason
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

12.4 Operational laws and consequences

- Support law: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

12.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact phase and interference record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

12.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.

- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Interference is exact predecessor merging with phase labels; no complex exponential is assumed.	PASS

12.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: phase, interference.

Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

12.8 Exact evidence identities

- Source manifest: sha256 : e7c78294725fc5b8365f70ee9be63ba92165aa712a7a686ecd999e74a8db7393
- Complete-census certificate:
sha256 : 3125c54031dc65d7e5e4963aa4b031acf83c57e009e731ee188d7806d07b7684

- Derivation seal: sha256:da60087f0965652227de2f3c71b7584d22e2f16a2b794c4a3fa4152bc27d186a
- Independent implementation:
sha256:5b0728c303020c2658658987b7794ae08beb9b00f6e2f8c6b48c1cc5251db79b
- Independent certificate:
sha256:09d8b5b79e90ad34e20b35f399aafd7bc0c8948366962366431e8411290575a0
- External validation:
sha256:3d7e92436970f0002b6da43d04afbba1374c525fb6b98c43a99f1ab40c60d44f
- Model-admitted engine receipt:
sha256:c1c7e3f2c71ea7de6d801c397cff01ae8a005651deddccc7acd83bddc857f12
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-PHASE-INTERFERENCE-001-c1c7e3f2c71ea7de.json

13. Derivation 6: Fold entangling-composition law

Claim: SFT-QUANTUM-ENTANGLEMENT-001

13.1 Scientific necessity and exact theorem

Fold entangling-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-PHASE-INTERFERENCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

13.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for entanglement.

All finite entanglement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

13.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

13.4 Operational laws and consequences

- Support law: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

13.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact entanglement record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

13.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS

Control	Expected	Observed	Result
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.	PASS

13.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: entanglement, nonseparable state.

Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

13.8 Exact evidence identities

- Source manifest: sha256 : afb46ed17a896275cc6922c9f10115e35b92520aa8f0e38eb4bf67a7399cd1c8
- Complete-census certificate:
sha256 : 3dccccf9a2fd85f710d29322e1e2200163c35004bfa3d73c701a6520b7404c6d
- Derivation seal: sha256 : d7fe096c012e1467e6d8253f0d04d02de4bc66f76b479cca2190386dc4298cec
- Independent implementation:
sha256 : ebd4b53c6a0c5b77e9d0f9fc9f7f031c0d9f1f9a77ba9fd886f9153df0e39966
- Independent certificate:
sha256 : cfef4e7a31d71a6197bc757243682f2fa66e4ce558a1284d2d35c0fd76a7dc43
- External validation:
sha256 : 57031064581dc6356636b1f6f65f980795d963109bd56945287e91e13dfb9657
- Model-admitted engine receipt:
sha256 : 6bfa64c6f896c5f7eaa3a55872f31cd487923853353a3069ec7a839afa1d3d0d
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-ENTANGLEMENT-001-6bfa64c6f896c5f7.json

14. Derivation 7: Fold quantum measurement-semantics law

Claim: SFT-QUANTUM-MEASUREMENT-001

14.1 Scientific necessity and exact theorem

Fold quantum measurement-semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ENTANGLEMENT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

14.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for measurement.

All finite measurement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

14.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

14.4 Operational laws and consequences

- Support law: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

14.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact measurement record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

14.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.	PASS

14.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum measurement, collapse.

No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

14.8 Exact evidence identities

- Source manifest: sha256:13904de9dde6457d62a60404ddea441608fa18f0c92f149958bb94b8333e32cf
- Complete-census certificate:
sha256:2f837ffe8611aa00de3dde2d72e76ac5b16ec836ee79b879b5d3379af0046e71
- Derivation seal: sha256:7d99cd7ce254a64f69ce210b7b3883177227978c792faeeb846094c4856dc9fa
- Independent implementation:
sha256:09615ec981ffda13c0fd1ea5163e580708612ce2202d5b23d98d1c3b806d66c7
- Independent certificate:
sha256:59226eca7492c0aebb48b4ab677e9b24e1bcfc38d8438d37385b00ed1ab13d1f
- External validation:
sha256:0b403b2a032ee01b1b8bbc6ae03c9a26b51f564dd38d11f1bc0d775672d89026
- Model-admitted engine receipt:
sha256:e29a71f7dff97835e28850ee8d352ac80aa9ceabb4399e07105d892175776a06
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-MEASUREMENT-001-e29a71f7dff97835.json

15. Derivation 8: Fold reversible transformation and gate law

Claim: SFT-QUANTUM-GATE-001

15.1 Scientific necessity and exact theorem

Fold reversible transformation and gate law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-MEASUREMENT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001

- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

15.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum transformations and gates.

All finite quantum transformations and gates structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical gate fidelity remains an empirical engineering question.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

15.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.

Eliminated candidates	Exact first-failure reason
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

15.4 Operational laws and consequences

- Support law: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

15.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum transformations and gates record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

15.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-GATE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical gate fidelity remains an empirical engineering question.	PASS

15.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum gate, unitary transformation.

Physical gate fidelity remains an empirical engineering question.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical gate fidelity remains an empirical engineering question.

15.8 Exact evidence identities

- Source manifest: sha256:42bb13c0646eea4cd1dbb2822a4cc213980cbb6bf622d108985528f6b540117f
- Complete-census certificate:
sha256:9c9d3340df70fc9aa032104c6f83309f96ba6ce28721c8c3187e1dd29749e38e
- Derivation seal: sha256:d93e94a7c4c89f79904d0068dc37fb48db7040bb45533a02cb96f8c45cb617c1
- Independent implementation:
sha256:629fa6869c3efd87e733736338d25b83adf646fe360ce5210876449f86aa18d5
- Independent certificate:
sha256:8c737f9cb9e579c043660fde438e8405e4f6288e9d65f74e470be69d59549c89
- External validation:
sha256:9c8ee6c1d374120eab1601d55d5d859b2391a968447e023a1f818301f7f114b7
- Model-admitted engine receipt:
sha256:495704206fd6c701e1baf8d1ec0b4bcfd430a002aea0bb76453a091e01ccfdaf
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-GATE-001-495704206fd6c701.json

16. Derivation 9: Fold quantum circuit syntax and semantics law

Claim: SFT-QUANTUM-CIRCUIT-001

16.1 Scientific necessity and exact theorem

Fold quantum circuit syntax and semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-GATE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

16.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum circuits.

All finite quantum circuits structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This derives circuit semantics; hardware timing and noise are external.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

16.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

16.4 Operational laws and consequences

- Support law: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

16.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum circuits record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

16.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CIRCUIT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This derives circuit semantics; hardware timing and noise are external.	PASS

16.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum circuit, circuit semantics.

This derives circuit semantics; hardware timing and noise are external.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum

- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This derives circuit semantics; hardware timing and noise are external.

16.8 Exact evidence identities

- Source manifest: sha256:d4825331a10928d077303fda7e766c916140bf5346464aedd390634a321b7586
- Complete-census certificate:
sha256:23cba4f14d69066c4576001a3d04ea7f054118d48815312b06b97cb9d14db74c
- Derivation seal: sha256:8fb9825eeec2daff4511f6aaf81f866fe575a23d070bbfbbef9d4d5d59837990
- Independent implementation:
sha256:1d370c1f6d10c8b50b5f4ce8980a2557dead4382a9a3565990df366ab519fdd6
- Independent certificate:
sha256:613d70fcb5954bd320bae00fe784bb03726d6ab1b3cc0c287a6df616d4746618
- External validation:
sha256:0a859bd98ada50cc93ff7f774518d1fa57ec325f78cda38db2219c74eaa48ceb
- Model-admitted engine receipt:
sha256:63e751cf9ab382cdc5c4b5dfd0c6c2f01ad90659bf2c9189357f141475d9e8aa
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-CIRCUIT-001-63e751cf9ab382cd.json

17. Derivation 10: Fold quantum universality law

Claim: SFT-QUANTUM-UNIVERSALITY-001

17.1 Scientific necessity and exact theorem

Fold quantum universality law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CIRCUIT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

17.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase

and observation preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum universal computation.

All finite quantum universal computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Universality is exact for every generated description in the frozen grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

17.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

17.4 Operational laws and consequences

- Support law: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

17.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum universal computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

17.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.

- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Universality is exact for every generated description in the frozen grammar.	PASS

17.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal quantum computer, universal gate set.

Universality is exact for every generated description in the frozen grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Universality is exact for every generated description in the frozen grammar.

17.8 Exact evidence identities

- Source manifest: sha256:59bfa7dc2618cb23f1897460501e00e4fdd1f0a0b1fab31fc0ffc9e06de8c549
- Complete-census certificate:
sha256:4af4128a42fc2e6efbb404ece4464adde28e77475392686199b67aa7a00117be
- Derivation seal: sha256:7f480fe1655f32f7ff3f640e9e0f508c501d21b0b44430f24ff41fcc84d35c0a
- Independent implementation:
sha256:00dc7bb66a94f5a7f0ced9c5b08dea0b489c312288e3cc0bf250f86bc8a7d89f
- Independent certificate:
sha256:0d501674df0431ac2f24be59fafd160e2c225807ab1b000013632f09ddcd1214
- External validation:
sha256:43e4644556a5eb410b174b4cdc7d6993d965e7f1b5636773dd41e3c67ba148d1
- Model-admitted engine receipt:
sha256:28b6a73e02450c70bf3ee49f3fdcdf2e28b30c463e05218b99cdc8cda723e86a

- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-UNIVERSALITY-001-28b6a73e02450c70.json

18. Derivation 11: Fold quantum algorithm-family law

Claim: SFT-QUANTUM-ALGORITHMS-001

18.1 Scientific necessity and exact theorem

Fold quantum algorithm-family law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-UNIVERSALITY-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

18.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum algorithms.

All finite quantum algorithms structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Named speedups require their own exact problem family and resource proof.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

18.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

18.4 Operational laws and consequences

- Support law: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.

- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

18.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum algorithms record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

18.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Named speedups require their own exact problem family and resource proof.	PASS

18.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum algorithm, quantum speedup.

Named speedups require their own exact problem family and resource proof.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Named speedups require their own exact problem family and resource proof.

18.8 Exact evidence identities

- Source manifest: sha256:779315c063473730553e1b94b47294d97acd1023e130e685d2aee25fd1c2ef67
- Complete-census certificate:
sha256:63c84d1546a1f83c51991b85670ab300de70ad6532104733532a81dc371c7282
- Derivation seal: sha256:cb94fae7c67d50383d283b028c580568c6d7e1fafb4a919217df581529703a87
- Independent implementation:
sha256:89bb200d9b5c23e863521d13d47d326f8df6f7e4ace5605cd4e62929121a8065
- Independent certificate:
sha256:8eac37074eb7bcf0579cb97aedc91fa70d056d0828515070ae6df1f157b8d22c
- External validation:
sha256:fe1b4e628eb13b88f9e401532d3f00a562cb8264986a662d0716f9bf766d6409
- Model-admitted engine receipt:
sha256:a33b13375d76ebf039fcc5711c0810ab99ac0b9456295bc7881d76c94558a86b
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-ALGORITHMS-001-a33b13375d76ebf0.json

19. Derivation 12: Fold quantum complexity law

Claim: SFT-QUANTUM-COMPLEXITY-001

19.1 Scientific necessity and exact theorem

Fold quantum complexity law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ALGORITHMS-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001

- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

19.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum computational resources.

All finite quantum computational resources structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

19.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.

Eliminated candidates	Exact first-failure reason
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

19.4 Operational laws and consequences

- Support law: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

19.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum computational resources record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

19.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.

- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No arbitrary separation between conventional complexity classes is claimed without a separate certificate.	PASS

19.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum complexity, BQP.

No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

19.8 Exact evidence identities

- Source manifest: sha256:8eac4910a233e30d537d3cc6dd92cfa0af9524abcf1bb6dae3c57759bf0f57f
- Complete-census certificate:
sha256:43fdd77acc2f1d28c1d909f806c2bf5d1bfd03d84effc0c51387915d78f83c24

- Derivation seal: sha256:2dabda188246d3553502bbfb24cebd18a4513e169b7291aa5ecbf9ca5fd66fe9
- Independent implementation:
sha256:81b3d49b18170b958f74bfe4f4a5cbd47bb4b0712c750027a7554d3576523419
- Independent certificate:
sha256:94b58ef5a03dc1e832cdf2ee0a5158ca6d8d9626c6d514f939aee795a7d0eee6
- External validation:
sha256:15ad7c676afff75fe9d9d991b0843f388b9061ed86b67929008d295eef2a2cd1
- Model-admitted engine receipt:
sha256:d6307d961273c2a383fec3b919c56319b910bccb58dbbcf5402a2a76a9ca8b6
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-COMPLEXITY-001-d6307d961273c2a3.json

20. Derivation 13: Fold quantum communication law

Claim: SFT-QUANTUM-COMMUNICATION-001

20.1 Scientific necessity and exact theorem

Fold quantum communication law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-COMPLEXITY-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

20.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum communication.

All finite quantum communication structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical transmission capacity and security require registered media experiments.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

20.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

20.4 Operational laws and consequences

- Support law: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

20.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum communication record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

20.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS

Control	Expected	Observed	Result
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical transmission capacity and security require registered media experiments.	PASS

20.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum communication, teleportation.

Physical transmission capacity and security require registered media experiments.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical transmission capacity and security require registered media experiments.

20.8 Exact evidence identities

- Source manifest: sha256:8cbc3dcc0ee105c719a4c9019ba6f993d6bd51d32427286b18f5317b9a8ab96c
- Complete-census certificate:
sha256:288320813c5011753ac302d9c4d463a58e58623ae71d344625eea30eafcafe70
- Derivation seal: sha256:2b321b0b8c520885563b2c770bed8e0ec83e5aa0720a05a2f679125c5d1f63cc
- Independent implementation:
sha256:c98029cec76386f0bea9768e3536a822803d2c7fb1eb3e16616272c1e0e97bec
- Independent certificate:
sha256:b8460f43cb576317854272866636f450a0a837fd34929ae4b947d681dd95eb91
- External validation:
sha256:a91fd78dd1739e147a926a006477eb6fa4bef506490fedcedb86827c0b56fd91
- Model-admitted engine receipt:
sha256:01059b1cbe86b894ae6137de5625e1dfcd1cd640ac5634c0ca2260977993b97a
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-COMMUNICATION-001-01059b1cbe86b894.json

21. Derivation 14: Fold quantum coding law

Claim: SFT-QUANTUM-CODING-001

21.1 Scientific necessity and exact theorem

Fold quantum coding law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-COMMUNICATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

21.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum coding.

All finite quantum coding structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The code theorem applies only to the registered generated error family.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

21.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

21.4 Operational laws and consequences

- Support law: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

21.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum coding record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

21.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CODING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The code theorem applies only to the registered generated error family.	PASS

21.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum code, encoding.

The code theorem applies only to the registered generated error family.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The code theorem applies only to the registered generated error family.

21.8 Exact evidence identities

- Source manifest: sha256:85e7aabcdad3f77f050dd76d25a562d3156936ff78a453bbf1f62cfd528e8a8
- Complete-census certificate:
sha256:93f1b6cf418d33752789ee30da4e3691204d71062d6e49138d4a6f9a292996aa
- Derivation seal: sha256:ea64d9cb56689b01f32e45805b41e053214e1e665f68f83c9fac52f894205efb
- Independent implementation:
sha256:3e3936d7ac16b4f76240d81294e3dab60dc4110611c44429cea46df1444590f2
- Independent certificate:
sha256:736fdb0c5a526c7b542e4eaf8fdf98681aae185d448844b242b1f371688b7746
- External validation:
sha256:e7785a525808797f14dba72e1e20acbd6dbb176c7799c1e71afca1bca34b271f
- Model-admitted engine receipt:
sha256:b79afc51951dfbebb30cd70fba329965b700fa14a16fb45b5754568ec7407a3e
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-CODING-001-b79afc51951dfbebb.json

22. Derivation 15: Fold quantum error-correction law

Claim: SFT-QUANTUM-ERROR-CORRECTION-001

22.1 Scientific necessity and exact theorem

Fold quantum error-correction law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CODING-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

22.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum error correction.

All finite quantum error correction structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The completed census covers generated fault depths one, two and three and the successor law for every positive finite t .

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

22.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

22.4 Operational laws and consequences

- Support law: forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

22.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum error correction record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

22.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS

Control	Expected	Observed	Result
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.	PASS

22.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum error correction, repetition code.

The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

22.8 Exact evidence identities

- Source manifest: sha256:5ac864ccd27449cb7c1a756b5ed53c4094709160f00cf0e8f19e58f01a20a2d9
- Complete-census certificate:
sha256:9712241740820a883217d33d982426a39c21903950f917eafa9fefcd38d970389
- Derivation seal: sha256:61cc113c2d425a9f9c9d2e13bb1a983bd94e88220a6d91a4f1815d41b5059ee6
- Independent implementation:
sha256:f3e41dd58f96c808ce4cadab817f8a4ee86d6369ef8e6e43ddfaacc792071af9
- Independent certificate:
sha256:721e7c933eb6f760790d3ee15cd54171a46f8ffa24c67f5b59e62872b12fb16b
- External validation:
sha256:c5784b25becf3d48609c4a3447a3cff73cd2822418dd3800282d1948dbe6d3ef
- Model-admitted engine receipt:
sha256:a4791dcbf8753c32e8304c5d5e80f9bf7f21a7356b6adb9422ea60d5eaf59656
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-ERROR-CORRECTION-001-a4791dcbf8753c32.json

23. Derivation 16: Fold quantum fault-tolerance law

Claim: SFT-QUANTUM-FAULT-TOLERANCE-001

23.1 Scientific necessity and exact theorem

Fold quantum fault-tolerance law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ERROR-CORRECTION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

23.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for fault-tolerant quantum computation.

All finite fault-tolerant quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

23.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

23.4 Operational laws and consequences

- Support law: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

23.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact fault-tolerant quantum computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

23.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.	PASS

23.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: fault tolerance, threshold theorem.

No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

23.8 Exact evidence identities

- Source manifest: sha256:3260e928de52083476ff03669bb6321999c2e7042d32a55027e98040c2d16c4e
- Complete-census certificate:
sha256:1e9b2a7fc565525f5353d4590607e2f6d1450e18f5c3db873f9f8b61c229960f
- Derivation seal: sha256:1816e6b491362e7cc5b5d306e833227bdf765559d55693d0b18553e2911f1a57
- Independent implementation:
sha256:ef9893933f17387358f0d7512e05451d206c5ce7a11fe348bbc1052bd42ee0dd
- Independent certificate:
sha256:0a28a0751ef1e16a701eb0fdf84d8d4a2583c0e4314f9cd1d4badfd2e5a10c7a
- External validation:
sha256:367a4648636683bd04852d04c27eb6360aede4d53948a7002d5d0f6ea0a37ea2
- Model-admitted engine receipt:
sha256:9f2526c5bd9f1625a3cee65b0cba4e9a600ecdc9a8031ac51367e35c37308b16
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-FAULT-TOLERANCE-001-9f2526c5bd9f1625.json

24. Derivation 17: Fold quantum simulation law

Claim: SFT-QUANTUM-SIMULATION-001

24.1 Scientific necessity and exact theorem

Fold quantum simulation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-FAULT-TOLERANCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001

- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

24.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum simulation.

All finite quantum simulation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

24.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.

Eliminated candidates	Exact first-failure reason
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

24.4 Operational laws and consequences

- Support law: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

24.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum simulation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

24.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-SIMULATION-001; result: PASS.

- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-SIMULATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.	PASS

24.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum simulation, Hamiltonian simulation.

A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

24.8 Exact evidence identities

- Source manifest: sha256:a376c3321387216b559a5d2543ece347699b17421f174831037f93ee4012ba23
- Complete-census certificate:
sha256:0036a9974eaf22533d40dbeffc708a9b259140447ebb99dfd5406040e63c341b
- Derivation seal: sha256:bb1d2c15251c769fbcf1692edc85f6253cb6f103dbffe04680cedb4d492cbff8
- Independent implementation:
sha256:72f00440c8b0cfcaa26940b8aef2728b3e74896b435207b9b8f9426348064df9
- Independent certificate:
sha256:fcdb1c83f2a12bee60d57319cf49492523c03723aee2e0aafa22e3a88abff900

- External validation:
sha256:2e19399541f4035708283c1c043842f1c5b8ffef1330ec004109a14ff3fdfbd0
- Model-admitted engine receipt:
sha256:daf61c89738fb9f6370bc244c3a7538a3c4037ab408dfd5d15e3301c57e817a6
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-SIMULATION-001-daf61c89738fb9f6.json

25. Derivation 18: Fold quantum verification law

Claim: SFT-QUANTUM-VERIFICATION-001

25.1 Scientific necessity and exact theorem

Fold quantum verification law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-SIMULATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

25.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum verification.

All finite quantum verification structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Security and correctness are exact only for the registered prover and resource grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

25.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

25.4 Operational laws and consequences

- Support law: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

25.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum verification record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

25.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-VERIFICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Security and correctness are exact only for the registered prover and resource grammar.	PASS

25.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum verification, QMA.

Security and correctness are exact only for the registered prover and resource grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Security and correctness are exact only for the registered prover and resource grammar.

25.8 Exact evidence identities

- Source manifest: sha256:d17c41d57b8e1a861e03983b2a5a7808ec8756e28c7361cb95bb1f5a159635ee
- Complete-census certificate:
sha256:c54587141b8c1a489ee2b3d83d53548c435f0746abda0ab11358bbacb1fbe8ef
- Derivation seal: sha256:62b1704737e141ef52eab6b9687ec06ab6de072c1ba2dd8cc678880fe50e3ce0
- Independent implementation:
sha256:00bb831c0de419ea512cd89dd478e15c3804c348f3b094de0d7321006f9f408f
- Independent certificate:
sha256:fa884fd0ca86e8ae875b76054696f32cd5ef731fbbd1986b9a02b84140ece03a
- External validation:
sha256:753bc09d36d4c45faa75aa9f02e0780bc6c49a45126630495deb18d72c74cd77
- Model-admitted engine receipt:
sha256:6ca70e1a9487b428bc6e04e5e3b1a11907b79d1ed92f7d03a013a49380ecce54
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-VERIFICATION-001-6ca70e1a9487b428.json

26. Derivation 19: Fold quantum learning law

Claim: SFT-QUANTUM-LEARNING-001

26.1 Scientific necessity and exact theorem

Fold quantum learning law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-VERIFICATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

26.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum learning.

All finite quantum learning structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

26.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

26.4 Operational laws and consequences

- Support law: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

26.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum learning record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

26.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-LEARNING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.	PASS

26.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum learning, quantum machine learning.

No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

26.8 Exact evidence identities

- Source manifest: sha256:720947592f997269012cdac1186615d307e7b3a553f0ff0d1e42eac0daedd840
- Complete-census certificate:
sha256:4f049e2fe5f2c3499b0b6012f93625016627146fccbfc47dc3bc5dfcca22831c
- Derivation seal: sha256:1d75a66403b8bd208fad0ca8cdd0de260f54865c8eb4e1b1984a25888845b435
- Independent implementation:
sha256:de8da3d2eb9a1c0c7cf06c6f851c311161c5e967d182c53c1d2d4f61d54045e3
- Independent certificate:
sha256:301937de6bfe10c6fc95c62a92efa37c27d5866795d316932652ba6dd953bd5d
- External validation:
sha256:2b0d54edaf033b569004c5d6e52026adaf3d9d5e8ad1129ff7ec246db2cebb42
- Model-admitted engine receipt:
sha256:c4388e69ff7412fe09e53b7e4b2a4eda12168ff6665ca351758c185ac7e018b7
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-LEARNING-001-c4388e69ff7412fe.json

27. Derivation 20: Full operational classical-quantum correspondence law

Claim: SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

27.1 Scientific necessity and exact theorem

Full operational classical-quantum correspondence law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-LEARNING-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

27.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for classical and quantum operational correspondence.

All finite classical and quantum operational correspondence structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

27.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

27.4 Operational laws and consequences

- Support law: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

27.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact classical and quantum operational correspondence record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

27.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.	PASS

27.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classical-quantum correspondence, deferred measurement.

Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

27.8 Exact evidence identities

- Source manifest: sha256 : a784ae4b69af81c489a660162333bca6e55fb56d401856af1d8e14516e863595
- Complete-census certificate:
sha256 : 922645a8e17e57036ddadbb6d3a35bf1e0a4a691864cb54ddfac3fb3f0978b59
- Derivation seal: sha256 : d10d9d064e2a63254c2cae2d5d7ea26173427927650eb375fbf698f69cf61f86
- Independent implementation:
sha256 : 1cd0dd23ea4b5b984d9a1ebe768005d0c09258970fd241aa5a02215cfdae7d2b
- Independent certificate:
sha256 : 74d501babc031290bfc0b5a041bed885d7ccb65d700b20676a8260f75fd91519
- External validation:
sha256 : 4c2be0efb0edeb4392351b788cd58b9331c850577e49ae1b0699ad2ad420445f
- Model-admitted engine receipt:
sha256 : e1712589e1aba3ce6536a3ba33d599e06bdf793a335079d1a775975bc234b614
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001-e1712589e1aba3ce.json

28. Derivation 21: Limits of Fold quantum computation law

Claim: SFT-QUANTUM-LIMITS-001

28.1 Scientific necessity and exact theorem

Limits of Fold quantum computation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

28.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for limits of quantum computation.

All finite limits of quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

28.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

28.4 Operational laws and consequences

- Support law: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

28.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact limits of quantum computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

28.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-LIMITS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.	PASS

28.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: limits of quantum computing, quantum computability.

Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate

- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

28.8 Exact evidence identities

- Source manifest: sha256 : f111f69d04fe049625cc50c9378bfca7d08fd3c2e6cc82cfbc7151cf183223db
- Complete-census certificate:
sha256 : d9059ae5cf371742769379c802668b1bc83493f42cf636184e5357c3e3279f95
- Derivation seal: sha256 : 4abd56778c0176f6426706806e75fdff866dedbbe2e78345585c0f9d0c6f360f
- Independent implementation:
sha256 : f1ed114f18db0541db98fc2966784fabca2116c53921fb473b68c2e866a204c5
- Independent certificate:
sha256 : e018eb19348a191249d0fa680329d0cf71d1c2e1d62b0734732426f74afb2cf5
- External validation:
sha256 : 2eaff27d824980b4d9b2b6cfade5b6a24002915fd4faa1ba47c521331707c176
- Model-admitted engine receipt:
sha256 : 332e3b1f6df5725ae3e5cf97d9080daa26fd501e7138f1b9d1c7b4837469c545
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-LIMITS-001-332e3b1f6df5725a.json

29. Branch synthesis

The 21 laws establish one operational machine family in which classical and quantum modes share descriptions, traces, resources and verification. Quantum distinctions arise from complete branch support, held phase action, predecessor merging and nonfactorable composition; they are not imported as a separate mathematical universe.

30. Reproduction and invalidation

Run the repository-wide one-command verifier. A reviewer can invalidate a result by producing an omitted same-boundary coordinate, a second survivor, a nonminimal form, an induction counterexample, a failed inverse, an incomplete observation record, an uncorrected registered error mask or a mismatched receipt.

31. Scope and next branch

The next branch is Physics. It must test whether independently derived Fold relations correspond to mechanics, fields, spacetime, thermodynamics, quantum phenomena, gravitation, matter, waves, fluids, plasmas and condensed structures. Formal quantum computation cannot select the natural laws; sealed observation must do that work.

References

1. Landauer R. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*. 1961;5:183-191. doi:10.1147/rd.53.0183.
2. Bennett CH. Logical reversibility of computation. *IBM Journal of Research and Development*. 1973;17:525-532. doi:10.1147/rd.176.0525.
3. Feynman RP. Simulating physics with computers. *International Journal of Theoretical Physics*. 1982;21:467-488. doi:10.1007/BF02650179.
4. Deutsch D. Quantum theory, the Church-Turing principle and the universal quantum computer. *Proceedings of the Royal Society A*. 1985;400:97-117. doi:10.1098/rspa.1985.0070.

5. Shor PW. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*. 1997;26:1484-1509. doi:10.1137/S0097539795293172.
6. Grover LK. A fast quantum mechanical algorithm for database search. *Proceedings of STOC*. 1996:212-219. doi:10.1145/237814.237866.
7. Schumacher B. Quantum coding. *Physical Review A*. 1995;51:2738-2747. doi:10.1103/PhysRevA.51.2738.
8. Smith M. Smithian Fold Theory Scientific Constitution. V3 clean-room repository, 2026.